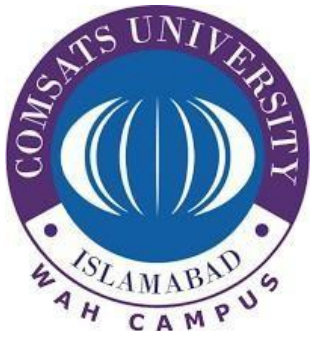


DEVOPS FOR CLOUD COMPUTING

Assignment

Name:
Tehreem Mirza

Registeration No:
FA22-BSE-088

Section:
BSE-8B

Submitted To:
Sir Muhammad Ibrahim

JDK 21 Download:

JDK 26 JDK 25 JDK 21

Java SE Development Kit 21.0.10 downloads

JDK 21 binaries are free to use in production and free to redistribute, at no cost, under the Oracle No-Fee Terms and Conditions (NFTC).

JDK 21 will receive updates under the NFTC, until September 2026, a year after the release of the next LTS. Subsequent JDK 21 updates will be licensed under the Java SE OTN License (OT production use beyond the limited free grants of the OTN license will require a fee.

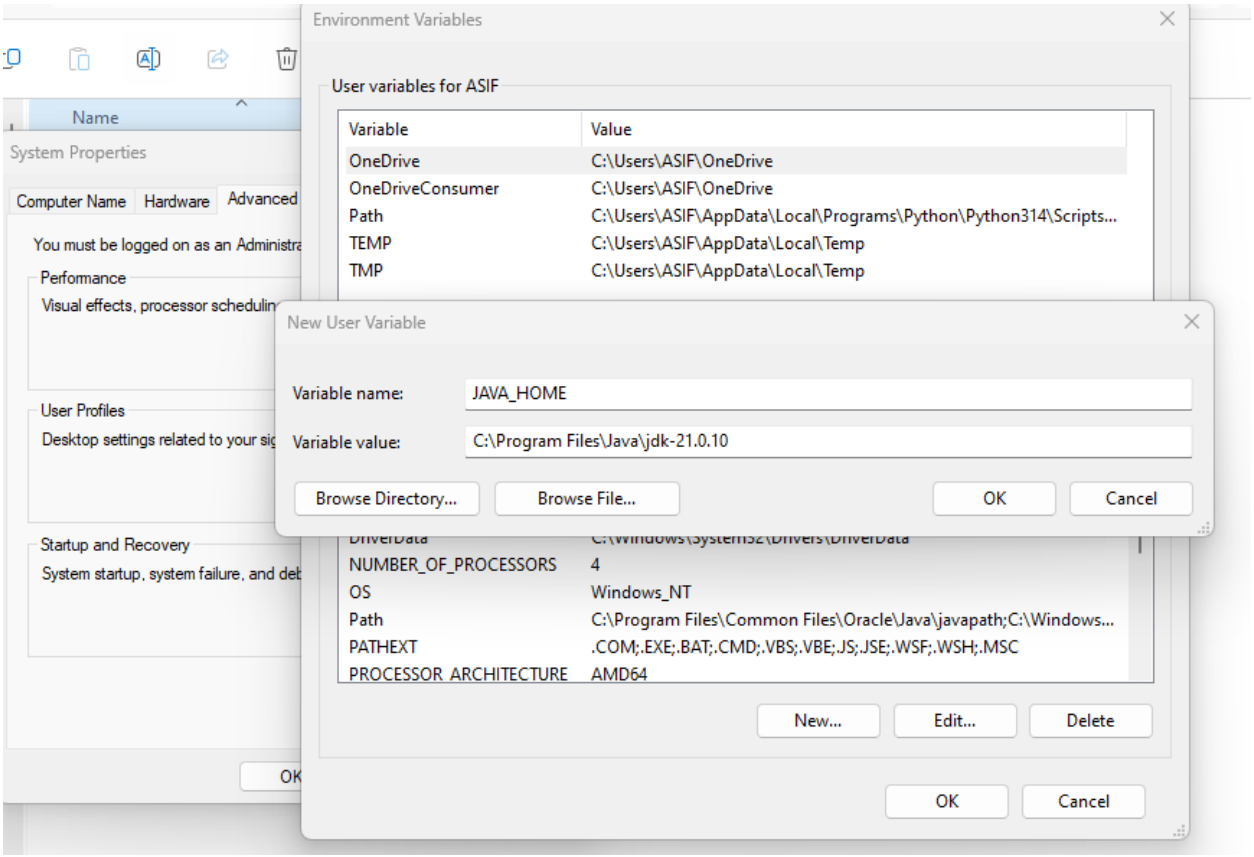
Linux macOS Windows

Product/file description	File size	Download
x64 Compressed Archive	186.54 MB	https://download.oracle.com/java/21/latest/jdk-21_windows-x64_bin.zip (sha256)
x64 Installer	165.68 MB	https://download.oracle.com/java/21/latest/jdk-21_windows-x64_bin.exe (sha256)

Activate Windows



Created a new variable in environmental variables



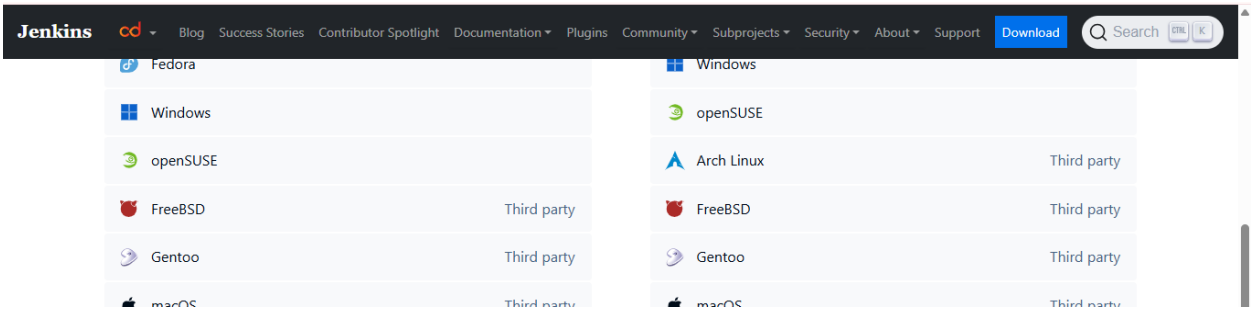
Java JDK installation is completed.

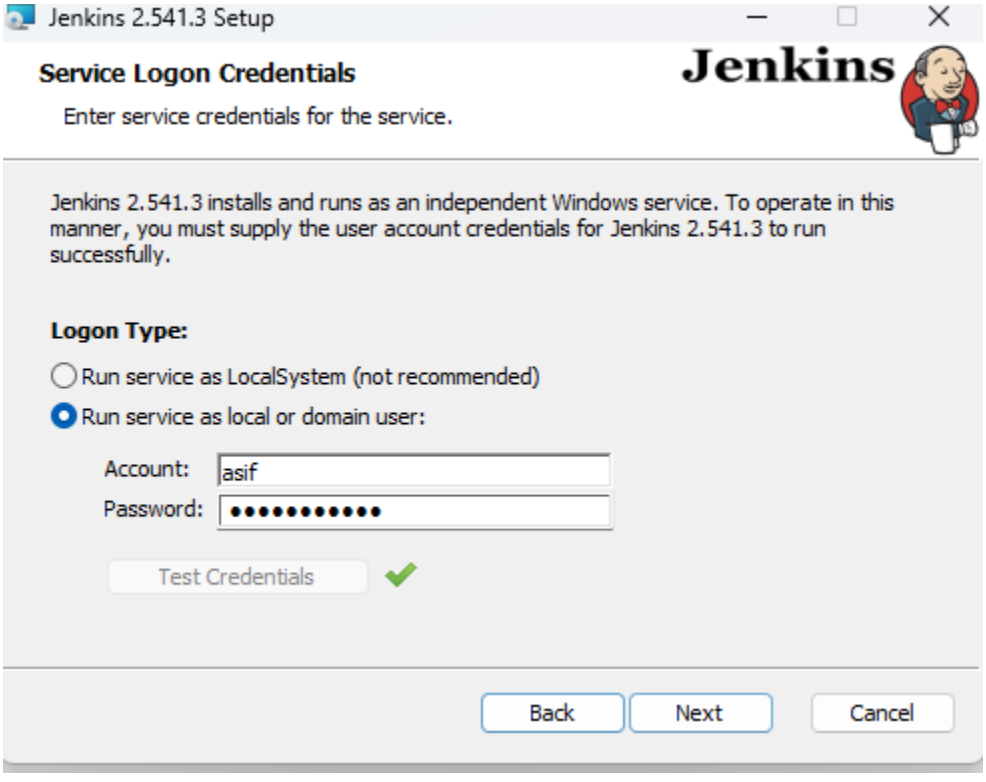
```
C:\Users\ASIF>java --version
java 21.0.10 2026-01-20 LTS
Java(TM) SE Runtime Environment (build 21.0.10+8-LTS-217)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.10+8-LTS-217, mixed mode, sharing)

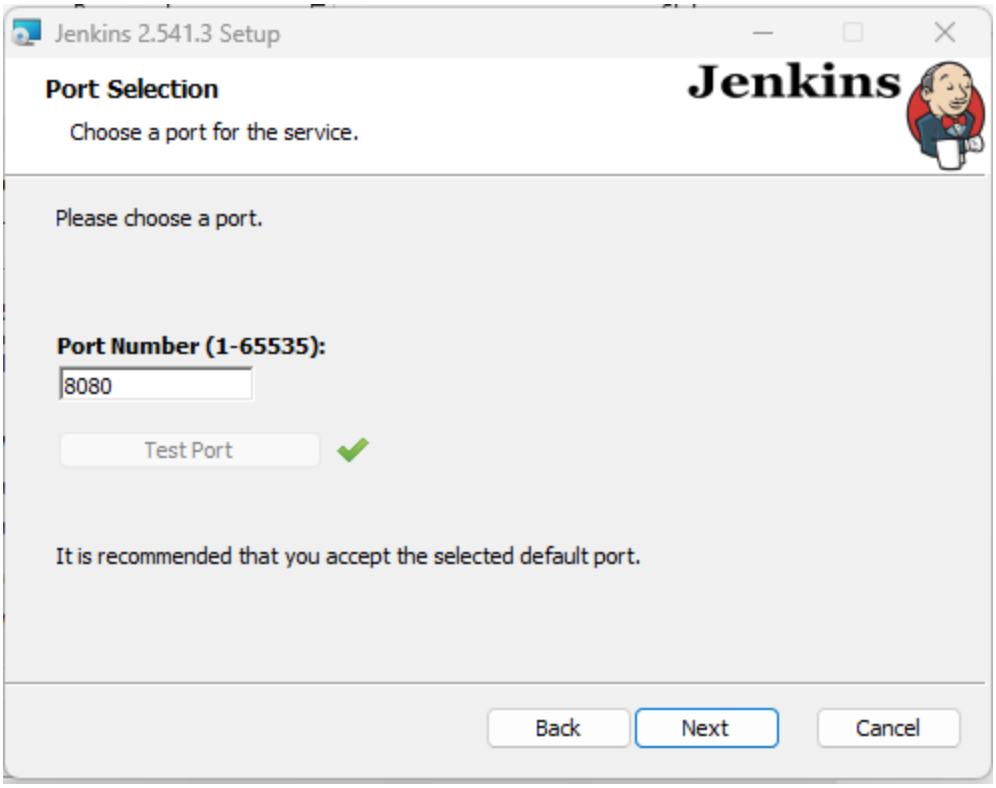
C:\Users\ASIF>
```

Installing Jenkins on windows

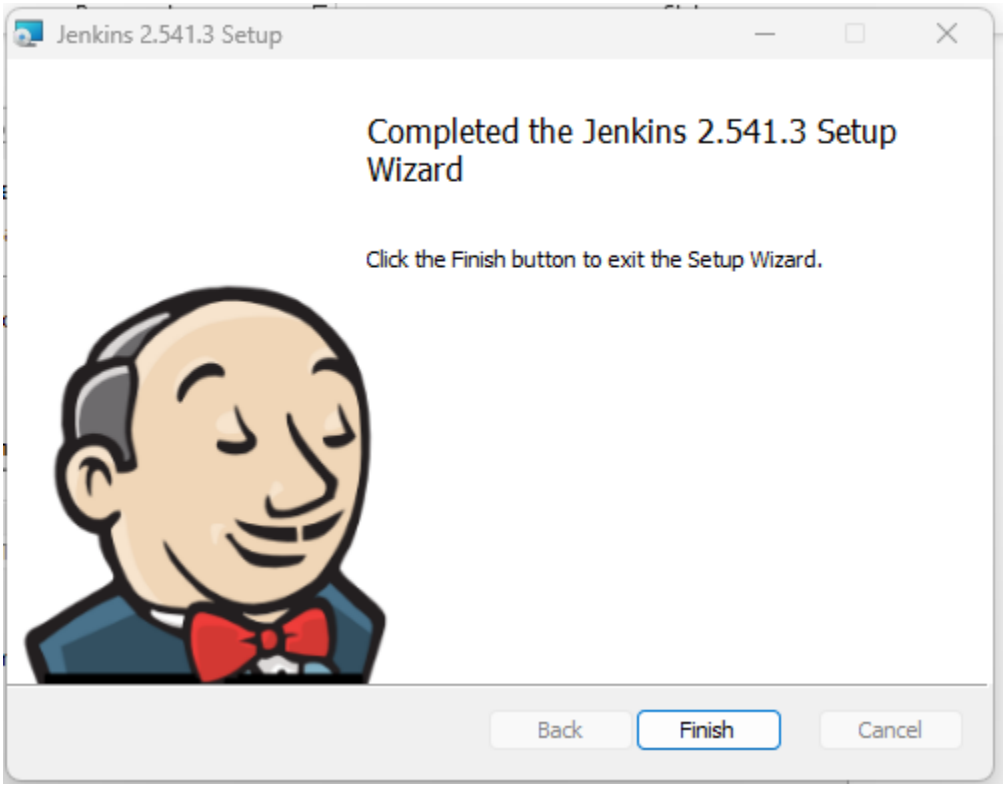
Open website and downloading windows one



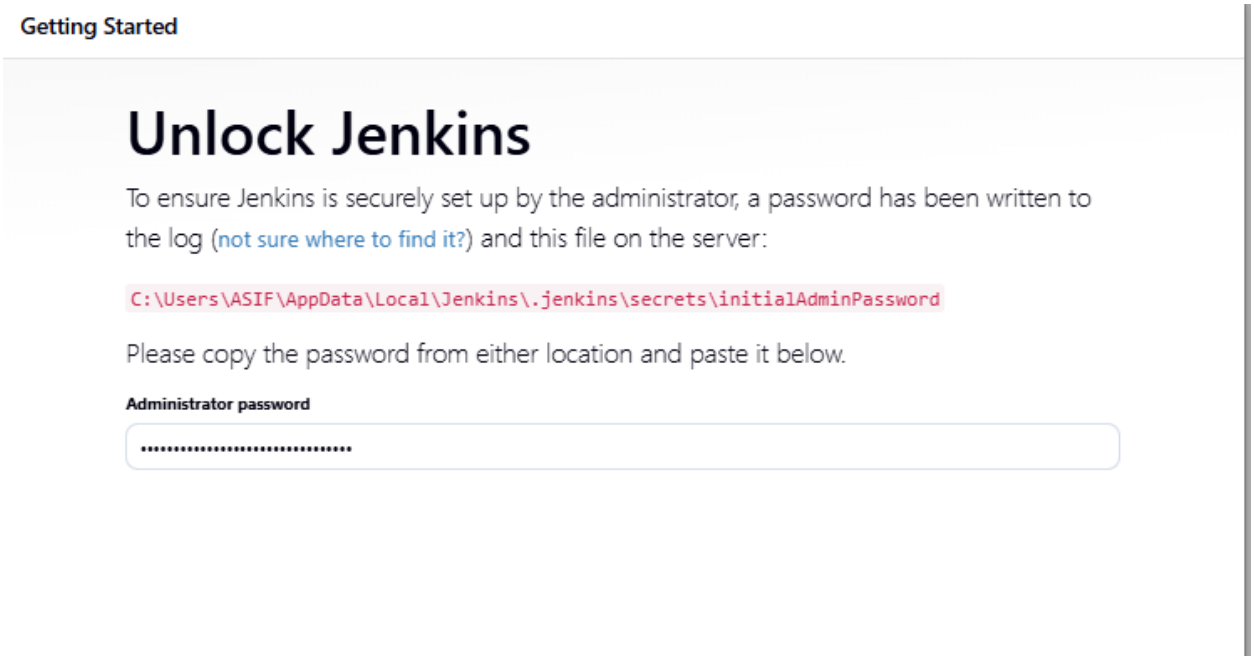



Jenkins 2.541.3 Setup
Port Selection
Choose a port for the service.
Jenkins
Please choose a port.
Port Number (1-65535):
8080
Test Port
It is recommended that you accept the selected default port.
Back
Next
Cancel

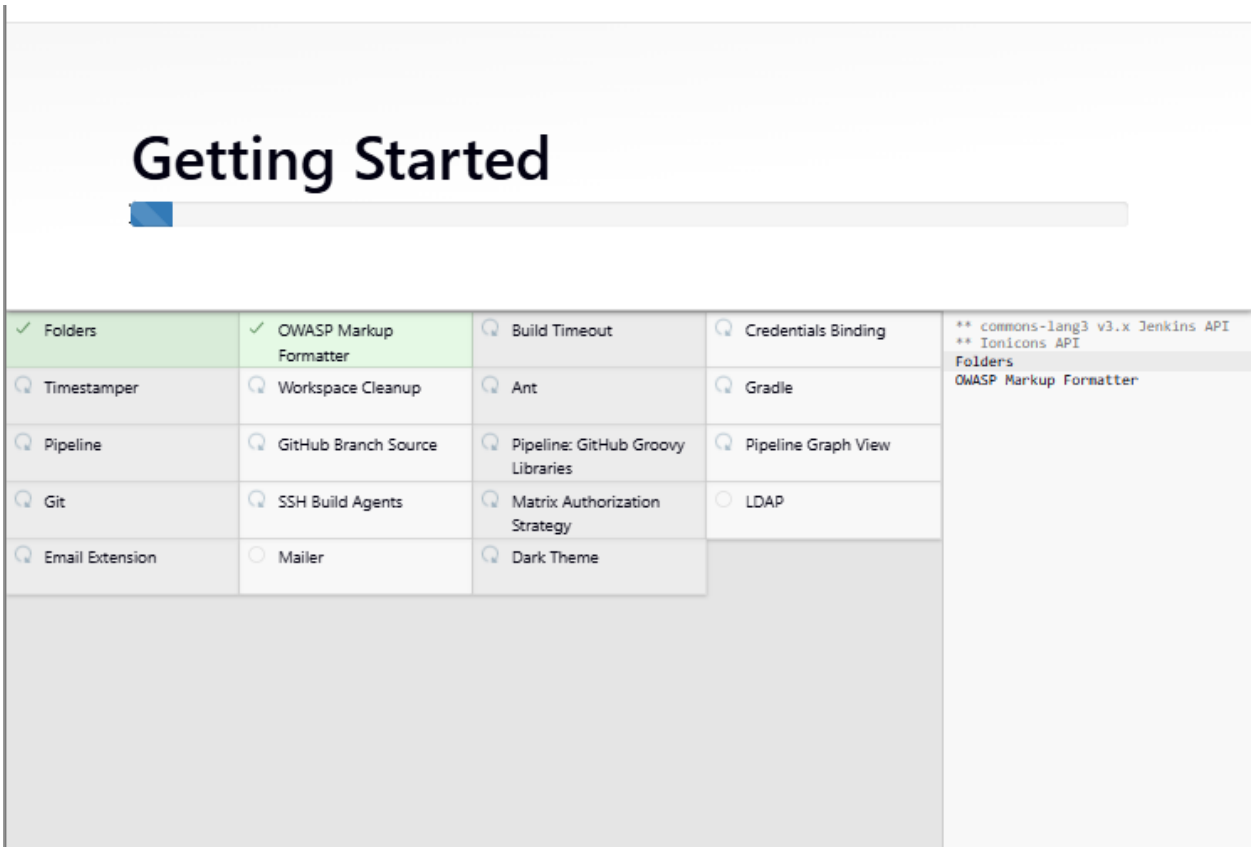
Jenkins installation completed

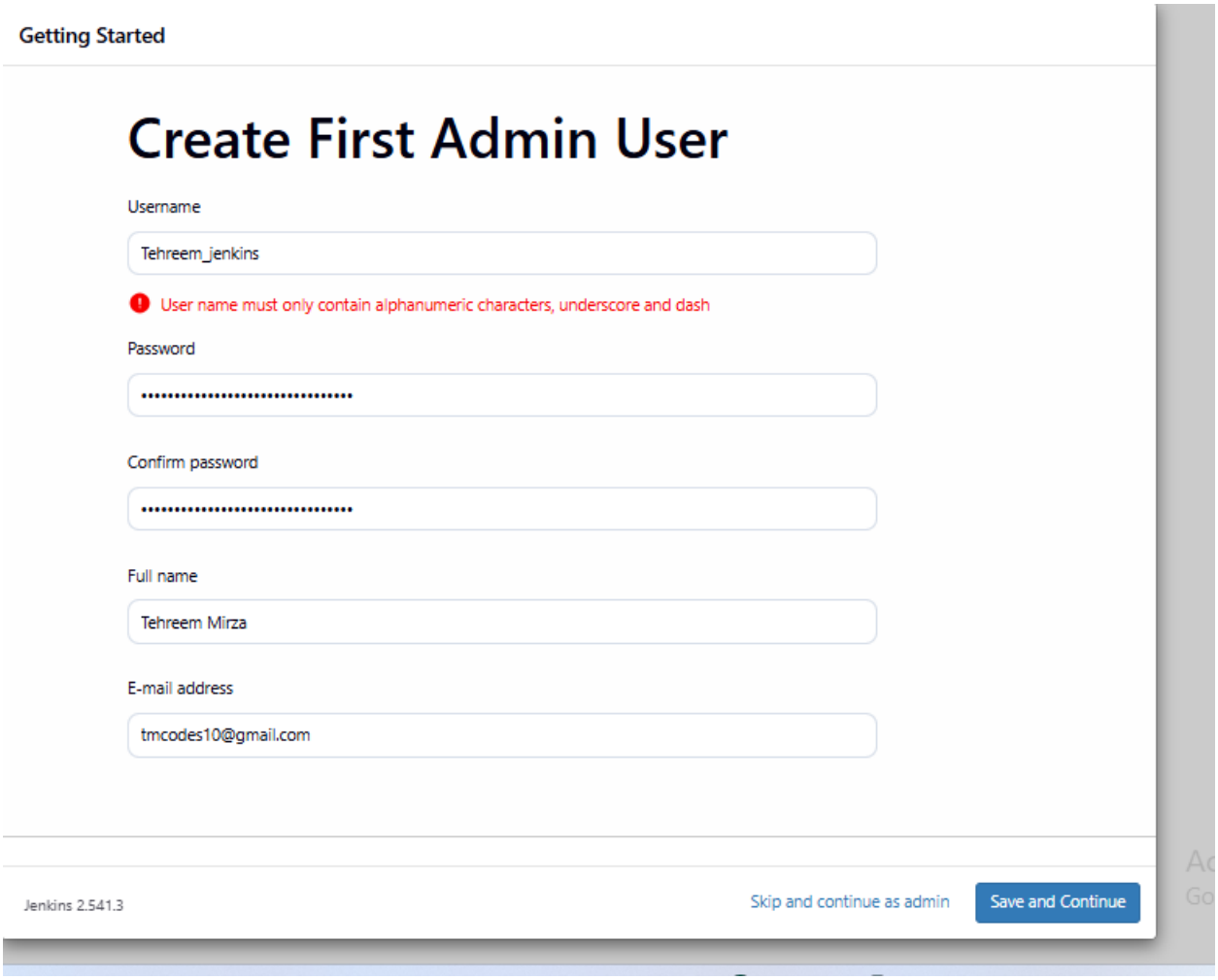


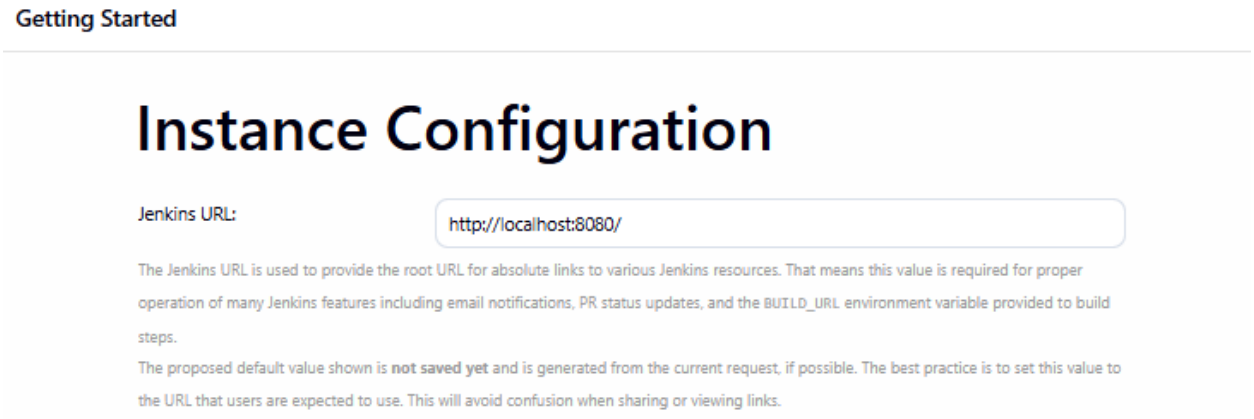
Now open localhost:8080



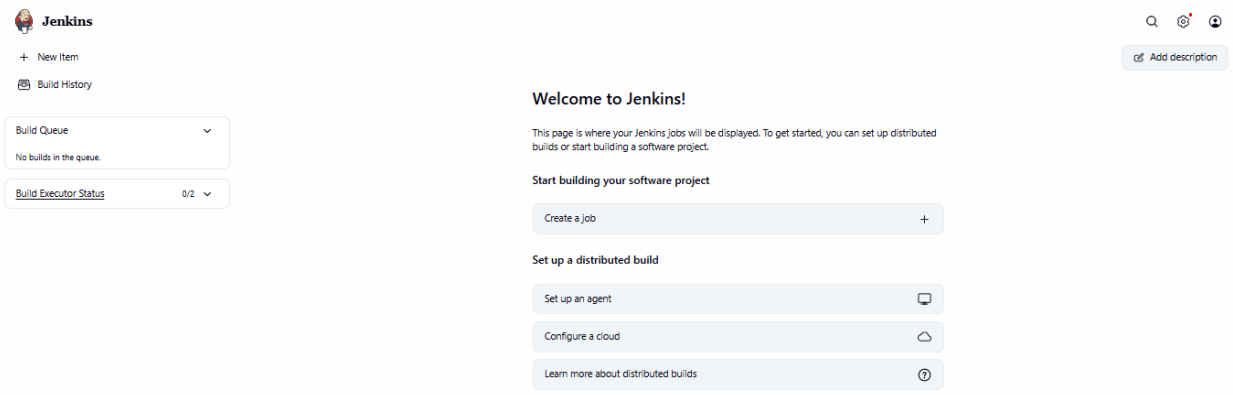
Installation of plugins

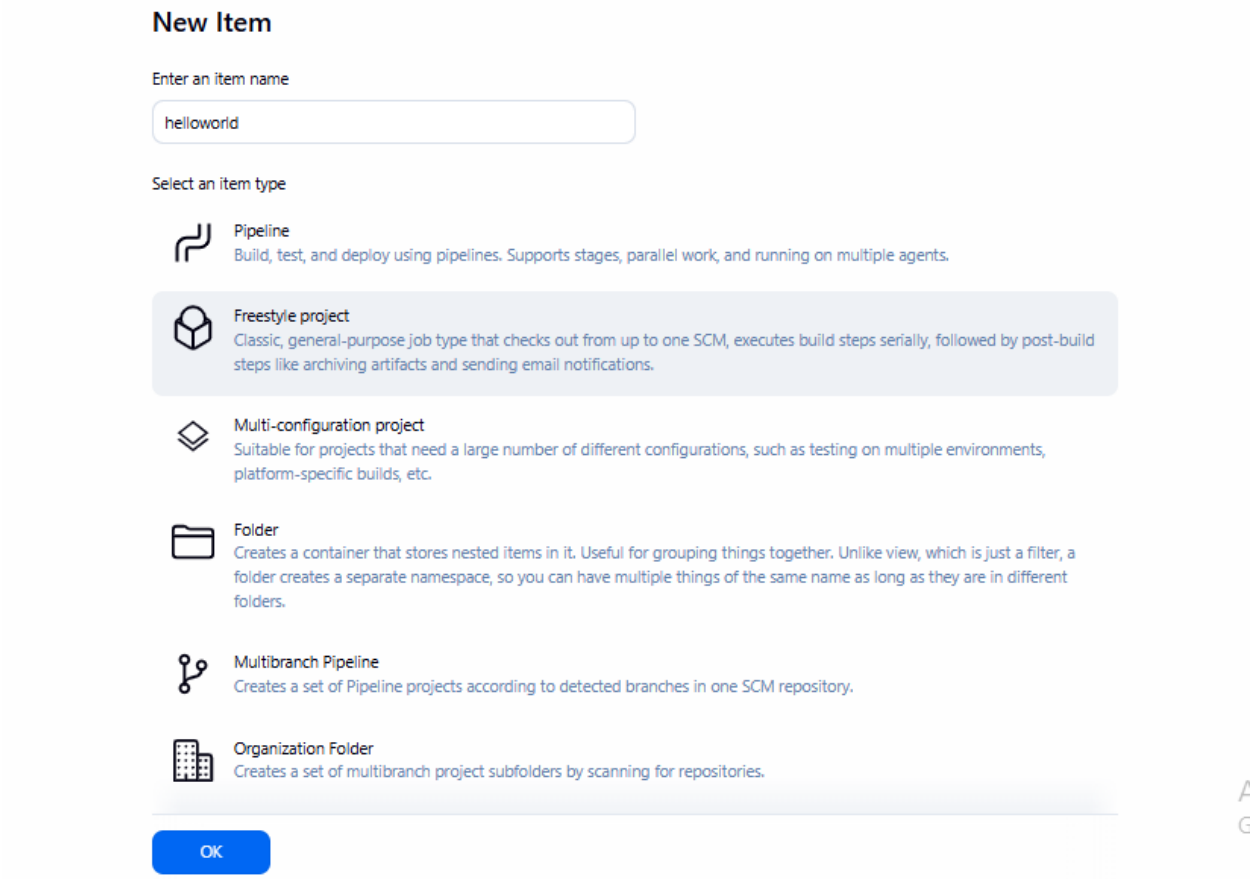


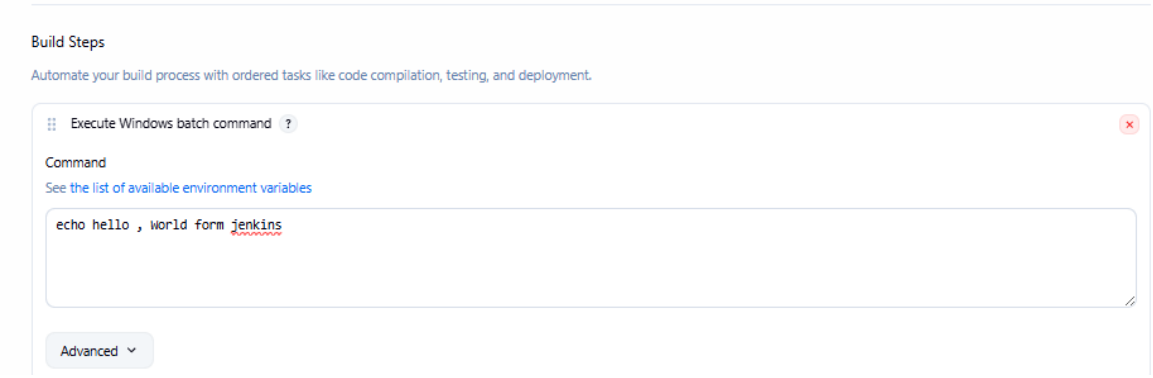

Getting Started
Create First Admin User
Username
Tehreem_jenkins
User name must only contain alphanumeric characters, underscore and dash
Password
Confirm password
Full name
Tehreem Mirza
E-mail address
tmcodes10@gmail.com
Jenkins 2.541.3
Skip and continue as admin
Save and Continue


Getting Started
Instance Configuration
Jenkins URL:
http://localhost:8080/
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.
The proposed default value shown is not saved yet and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

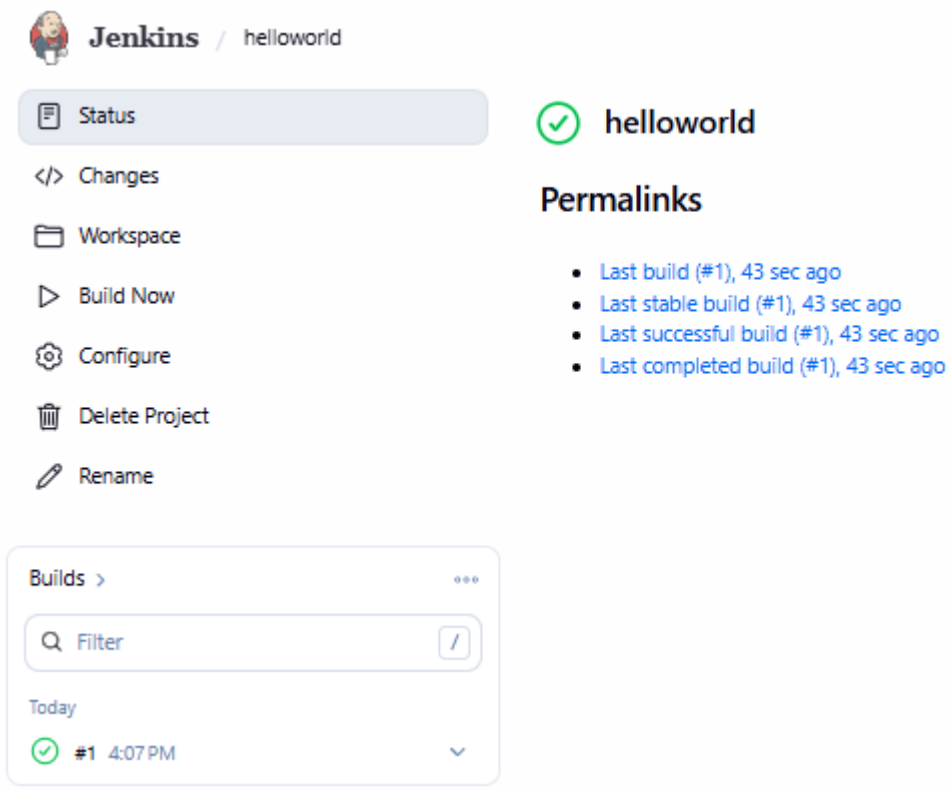
Jenkins is now installed and completed





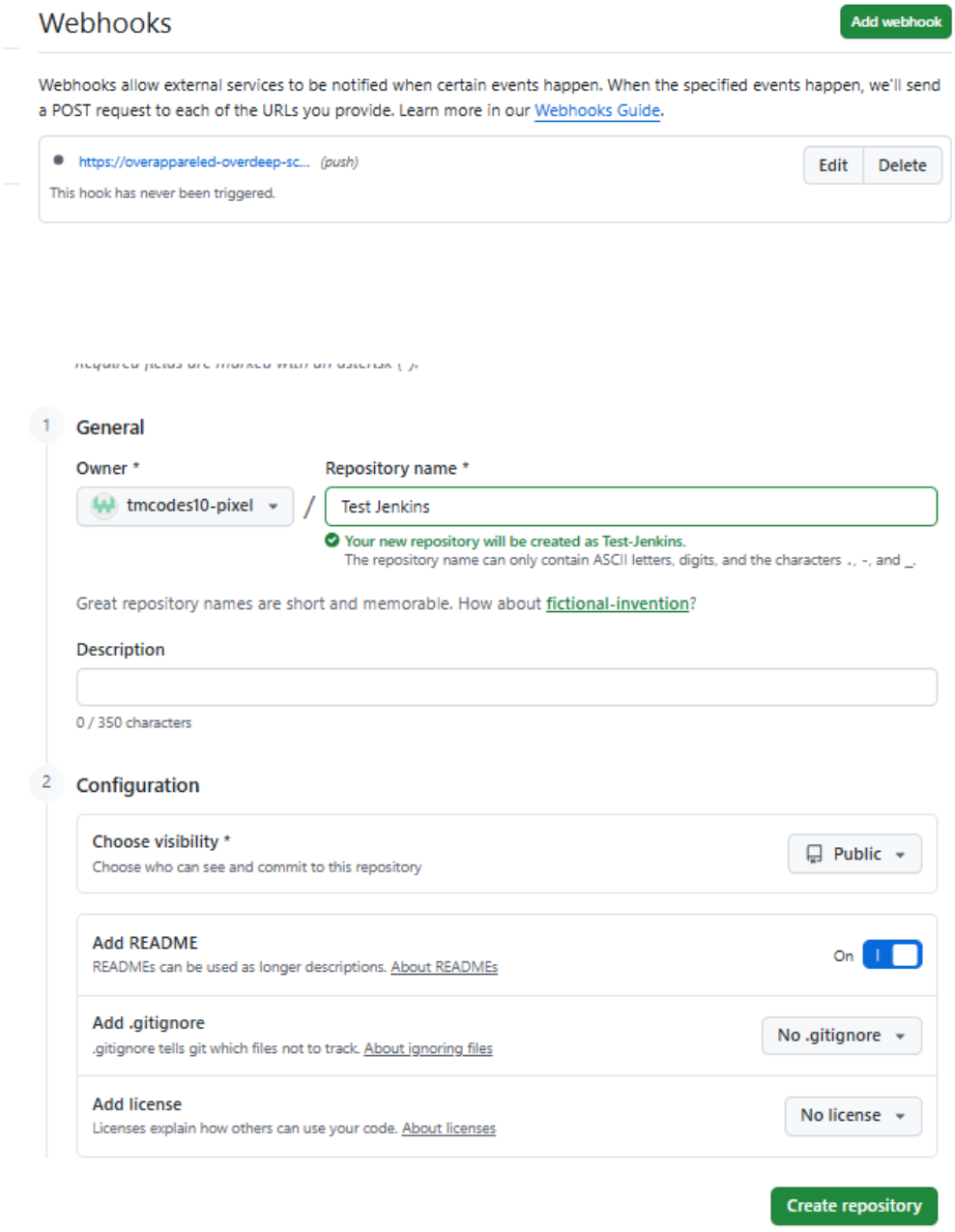


Simple hello world program run



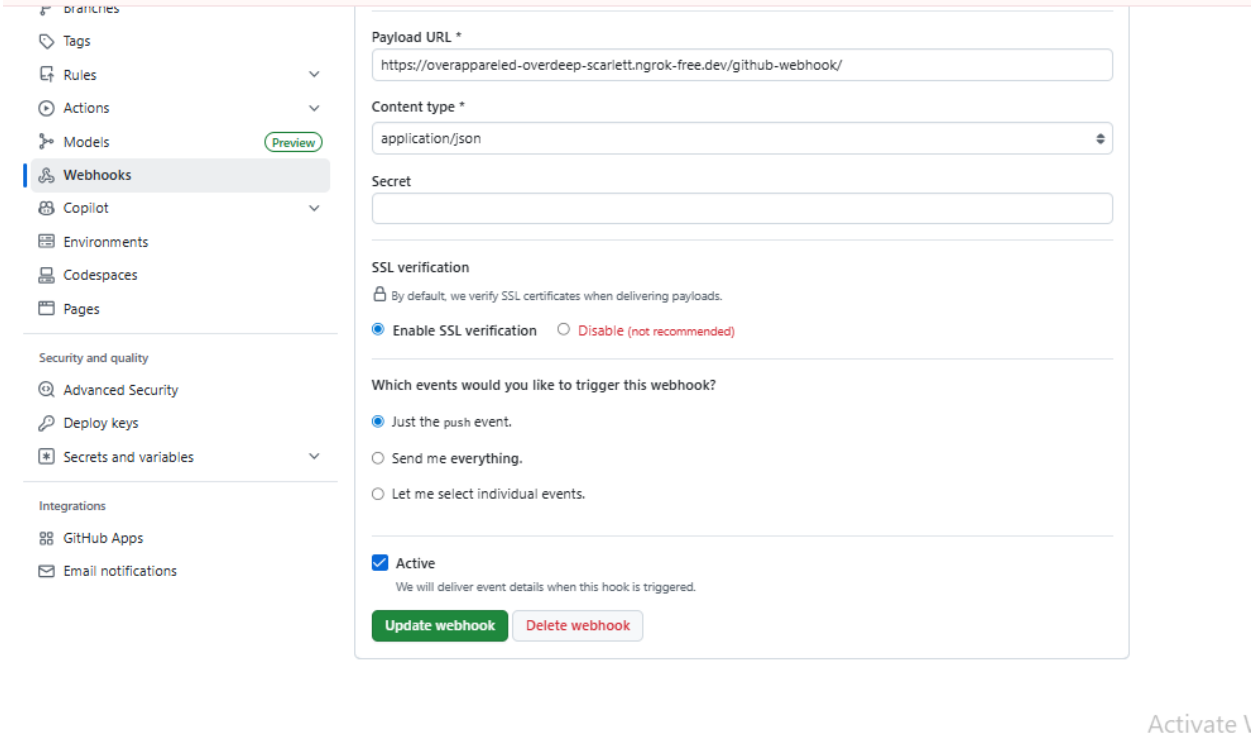
GitHub connection

Creating repository on github

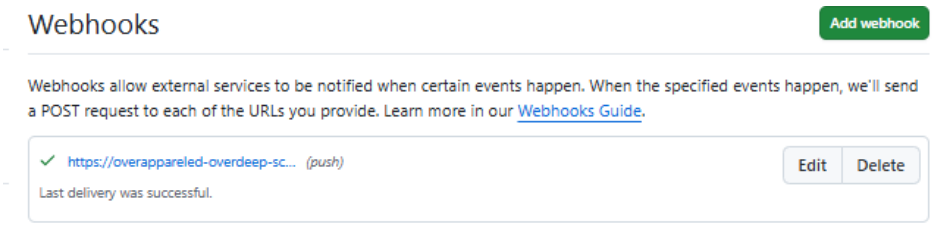


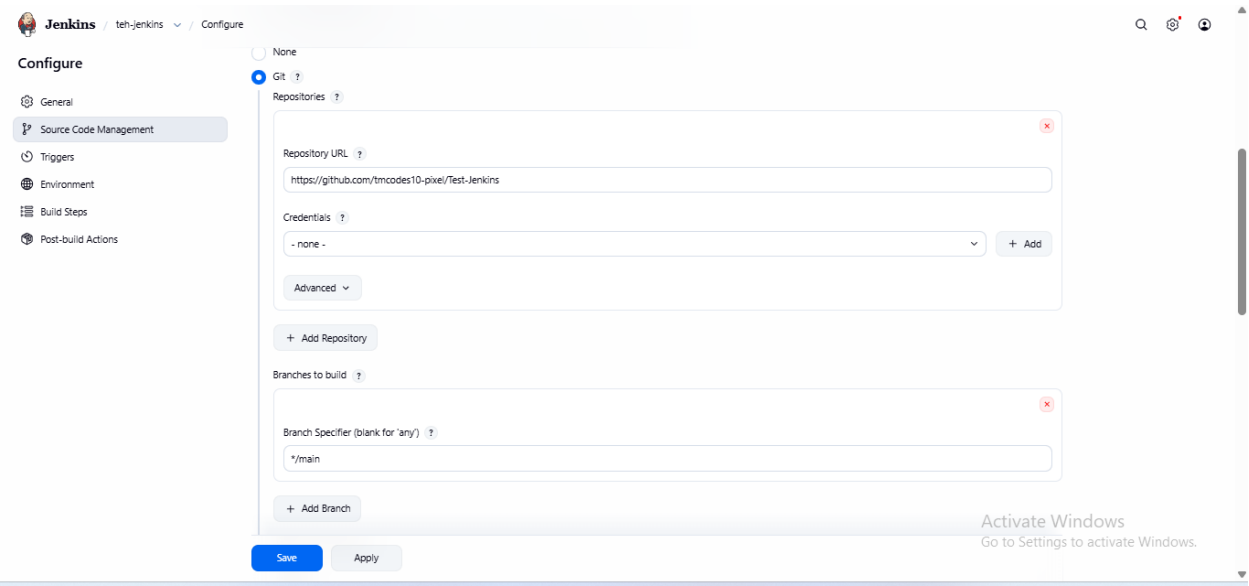
Activat
Go to Set

Adding ngrok link to webhook

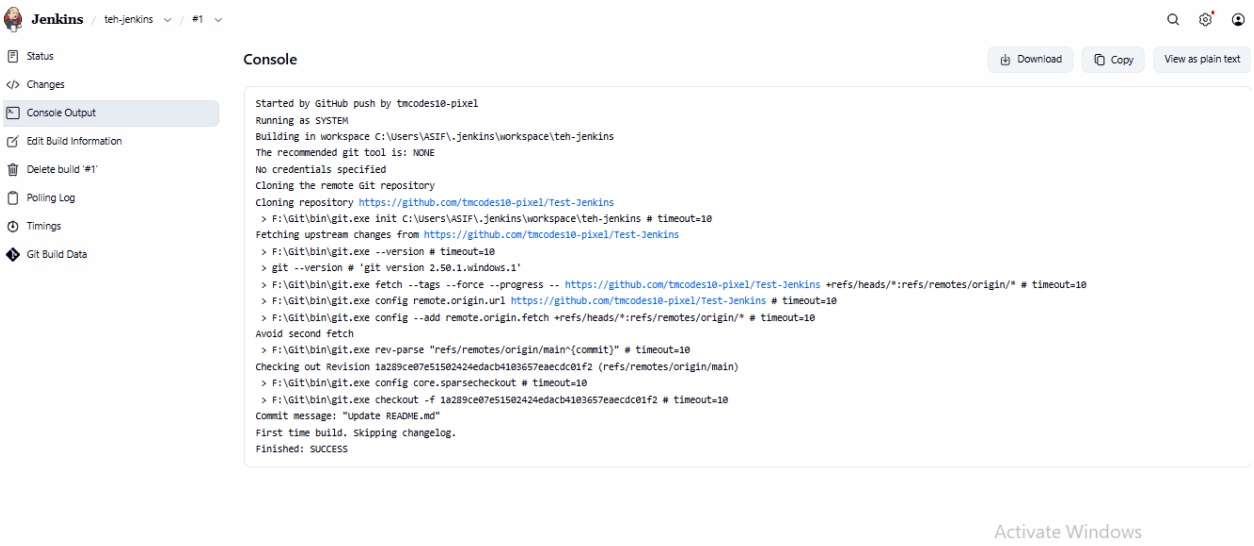


Activate \




Jenkins / teh-jenkins / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
None
Git
Repositories
Repository URL
https://github.com/tmcodes10-pixel/Test-Jenkins
Credentials
- none -
Add
Advanced
Add Repository
Branches to build
Branch Specifier (blank for 'any')
*/main
Add Branch
Save
Apply
Activate Windows
Go to Settings to activate Windows.


Triggers
Set up automated actions that start your build based on specific events, like code changes or scheduled times.
Trigger builds remotely (e.g., from scripts)
Build after other projects are built
Build periodically
GitHub hook trigger for GITScm polling
Poll SCM


Jenkins
teh-jenkins
#1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Polling Log
Timings
Git Build Data
Console
Download
Copy
View as plain text
Started by GitHub push by tmcodes10-pixel
Running as SYSTEM
Building in workspace C:\Users\ASIF\.jenkins\workspace\teh-jenkins
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/tmcodes10-pixel/Test-Jenkins
> F:\Git\bin\git.exe init C:\Users\ASIF\.jenkins\workspace\teh-jenkins # timeout=10
Fetching upstream changes from https://github.com/tmcodes10-pixel/Test-Jenkins
> F:\Git\bin\git.exe --version # timeout=10
> git --version # 'git version 2.50.1.windows.1'
> F:\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/tmcodes10-pixel/Test-Jenkins +refs/heads/*:refs/remotes/origin/* # timeout=10
> F:\Git\bin\git.exe config remote.origin.url https://github.com/tmcodes10-pixel/Test-Jenkins # timeout=10
> F:\Git\bin\git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> F:\Git\bin\git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 1a289ce07e51502424edacb4103657eaecdc01f2 (refs/remotes/origin/main)
> F:\Git\bin\git.exe config core.sparsecheckout # timeout=10
> F:\Git\bin\git.exe checkout -f 1a289ce07e51502424edacb4103657eaecdc01f2 # timeout=10
Commit message: "Update README.md"
First time build. Skipping changelog.
Finished: SUCCESS
Activate Windows